

NETWORK CABLING

A Simple & Complete Guide

Covers: LAN Cables · Optical Fiber · SFP Modules

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Chapter 1: What is a Network Cable?

A **network cable** is the physical wire that carries data signals between computers, switches, routers, and other network devices. Think of it like a road — data travels through the cable just like cars travel on a road.

There are **two main families** of network cables:

- **Copper cables** — carry electrical signals (cheap, common, used for LAN)
- **Fiber optic cables** — carry light signals (fast, long-distance)

■ **Tip:** Copper cables are perfect for within a building. Fiber cables are used between buildings or cities!

Chapter 2: LAN Copper Cable Types

2.1 UTP – Unshielded Twisted Pair (Most Common!)

UTP is the most widely used network cable in homes and offices. It has **4 pairs of copper wires twisted together** inside a plastic jacket. Twisting the wires reduces interference.

- No metal shielding — lighter and flexible
- Works great in homes, offices, schools
- Susceptible to electrical interference (motors, lights)
- Uses RJ-45 connectors (same as phone but 8 pins)

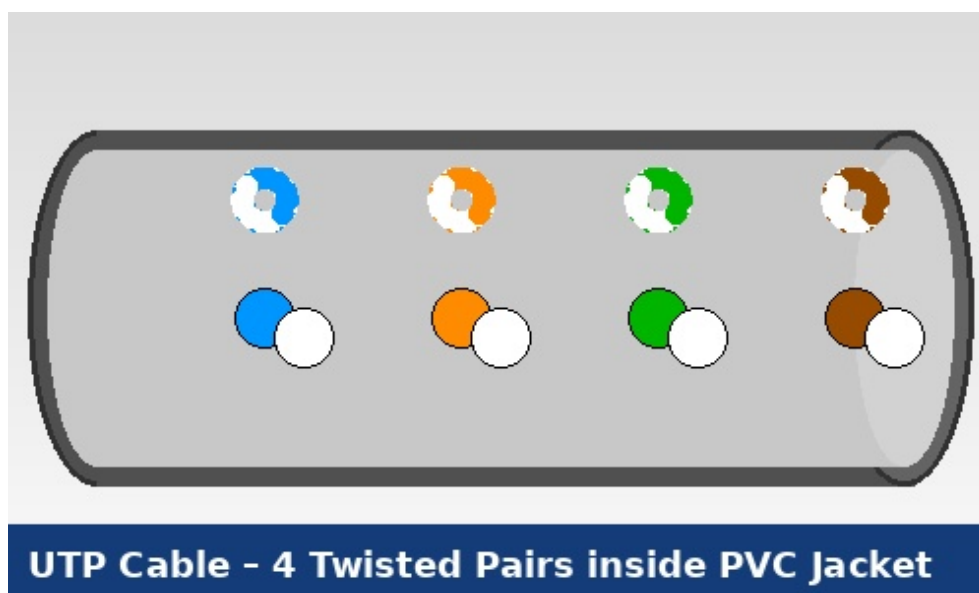


Fig 1: UTP Cable – 4 colour-coded twisted pairs inside a grey PVC jacket

2.2 STP – Shielded Twisted Pair

STP is similar to UTP but has a **metal foil or braid shield** wrapped around the wires. This shield blocks electrical interference from outside. Used in factories, hospitals, or anywhere there is lots of electrical noise.

- Better noise protection than UTP
- Heavier and stiffer — harder to install
- Must be properly grounded or it makes interference worse!
- Sub-types: FTP (foil), SFTP (foil+braid), SSTP (each pair shielded)

2.3 Coaxial Cable (Legacy)

Coaxial cable has a **single central copper wire** surrounded by insulation, a metal braid shield, and an outer jacket. It was used in older Ethernet networks (10BASE-2, 10BASE-5) and is still used for cable TV and CCTV cameras.

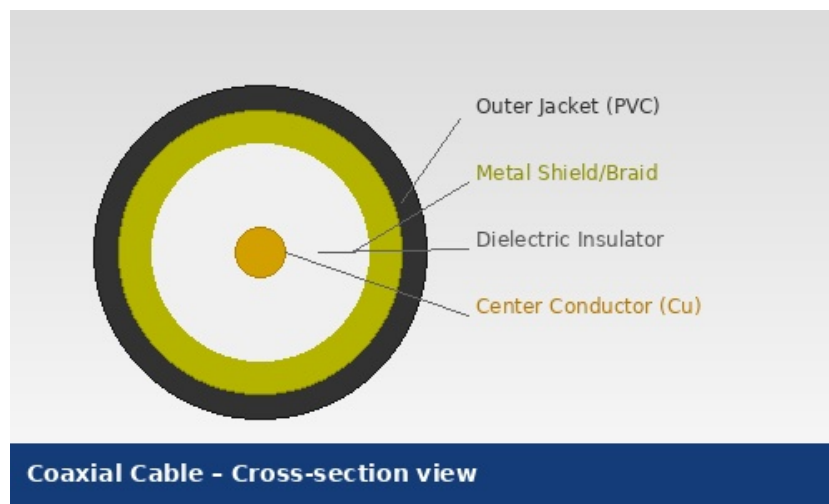


Fig 2: Coaxial Cable Cross-Section – Centre conductor, dielectric, braid shield, PVC jacket

■ **Tip:** Coaxial cables are no longer used for modern Ethernet networks — they have been replaced by UTP!

Chapter 3: Cable Categories – Which One Should I Use?

Each category (Cat) of UTP cable supports different **speeds and bandwidths**. Higher categories support faster speeds. The cable category is printed on its jacket.

Category	Speed	Bandwidth	Max Length	Best For
Cat 5	100 Mbps	100 MHz	100 m	Legacy – do not use
Cat 5e	1 Gbps	100 MHz	100 m	Home / small office
Cat 6	1 Gbps	250 MHz	100 m	Office – reduces crosstalk
Cat 6a	10 Gbps	500 MHz	100 m	★ Recommended new install
Cat 7	10 Gbps	600 MHz	100 m	Industrial / shielded
Cat 8	25–40 Gbps	2000 MHz	30 m	Data centre server racks

■ **Tip:** Cat 6a is today's recommended choice for any new installation — it supports 10 Gbps and is future-proof!

Simple Rule to Remember:

- All categories use the same **RJ-45 plug** — they are compatible!
- Higher category cable works on slower equipment (backward compatible)
- Maximum cable run is **100 metres** for all categories (except Cat 8 = 30 m)
- PoE (Power over Ethernet) also travels through the cable — use Cat 6+ for best results

Chapter 4: The RJ-45 Connector & Wiring

The **RJ-45** is the standard plug/socket used on LAN cables. It looks like a wider version of a telephone connector. It has **8 metal pins** that make contact with the 8 wires.

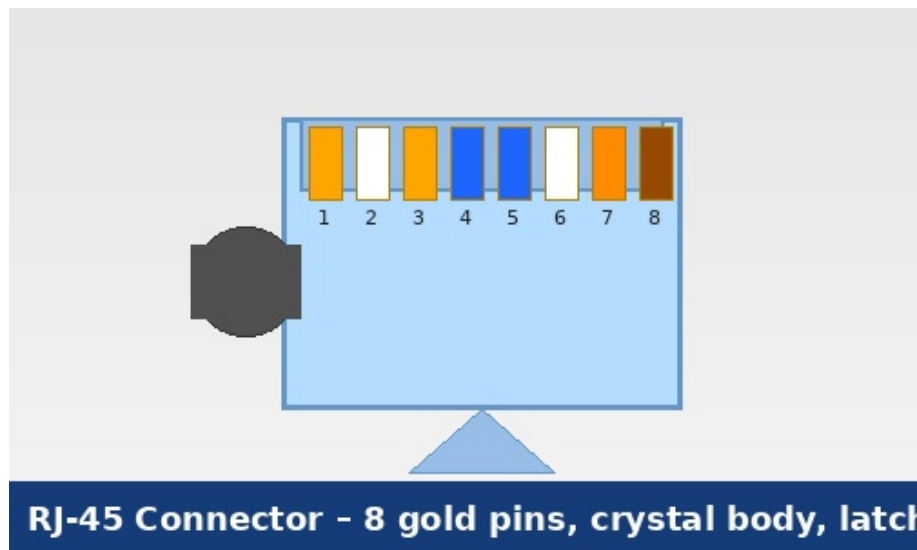


Fig 3: RJ-45 Connector – 8 gold pins, clear crystal body, latch tab underneath

T568A vs T568B – Two Wiring Standards

There are two ways to connect the 8 wires to the 8 pins. **T568B** is more common in the USA; **T568A** is used in Europe and government installations. You **MUST** use the same standard on both ends!

Pin	T568A (wire colour)	T568B (wire colour)	Signal
1	White / Green	White / Orange	TX+ Send
2	Green	Orange	TX- Send
3	White / Orange	White / Green	RX+ Receive
4	Blue	Blue	Unused / PoE
5	White / Blue	White / Blue	Unused / PoE
6	Orange	Green	RX- Receive
7	White / Brown	White / Brown	Unused / PoE
8	Brown	Brown	Unused / PoE

Cable Type	Standard Used	When to Use
Straight-Through	T568B → T568B	PC to Switch, Router to Switch (most common)
Crossover	T568A → T568B	PC to PC, Switch to Switch (legacy — rare today)

■ **Tip:** Modern switches and PCs have Auto-MDI-X — they automatically detect the cable type. Crossover cables are almost never needed today!

Chapter 5: Optical Fiber Cable

Optical fiber cables carry data as **pulses of light** through a very thin glass or plastic thread. Because light travels so fast, fiber cables can send data much faster and much farther than copper cables.

How does it work?

- A laser or LED generates a light pulse
- The light travels through the glass **core** using a principle called Total Internal Reflection
- The light bounces off the inner wall of the glass and cannot escape — like a mirror tunnel
- A receiver at the other end detects the light pulses and converts them back to data

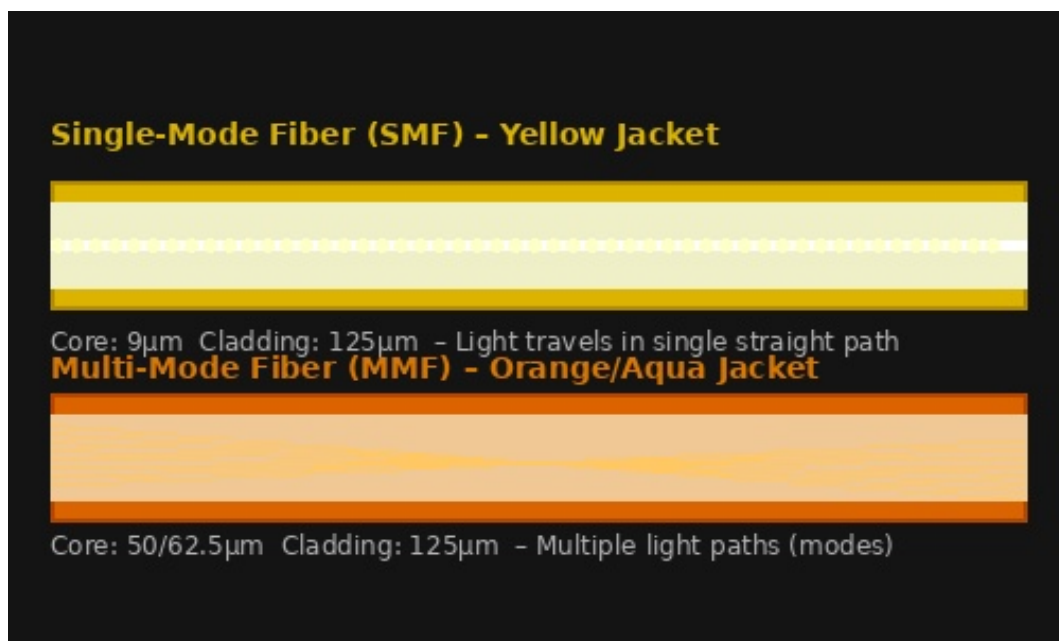


Fig 4: Single-Mode (yellow) vs Multi-Mode (orange) fiber — light travels through the core

5.1 Single-Mode Fiber (SMF) — Yellow Jacket

Single-mode fiber has a very **thin core (9 micrometres)** — about 100× thinner than a human hair! It allows only one beam of light to travel straight through.

- Speed: Extremely high — virtually unlimited bandwidth
- Distance: Up to 80 km (even more with amplifiers!)
- Cost: Higher — requires a laser transmitter
- Use: Long-distance links — city to city, building to building, WAN

5.2 Multi-Mode Fiber (MMF) — Orange or Aqua Jacket

Multi-mode fiber has a **larger core (50 or 62.5 micrometres)** that allows many beams of light to travel at slightly different angles at the same time.

- Speed: High — limited by distance due to light spreading
- Distance: Up to 300–400 m at 10 Gbps (depends on OM grade)
- Cost: Lower — uses cheaper LED or VCSEL transmitters
- Use: Inside a building — floor to floor, room to room

Type	Bandwidth	Core Size	Jacket Colour	10G Distance
OM1	200 MHz·km	62.5 µm	Orange	33 m
OM2	500 MHz·km	50 µm	Orange	82 m
OM3	2,000 MHz·km	50 µm	Aqua	300 m
OM4	4,700 MHz·km	50 µm	Aqua	400 m
OM5	28,000 MHz·km	50 µm	Lime Green	400 m

■ **Tip:** OM3 or OM4 (aqua) is the modern standard for data centres. OM1/OM2 (orange) is old — avoid for new installs!

Chapter 6: Fiber Optic Connectors

Unlike copper cables that all use RJ-45, fiber cables have **many different connector types**. The connector you need depends on the equipment and the cable type.

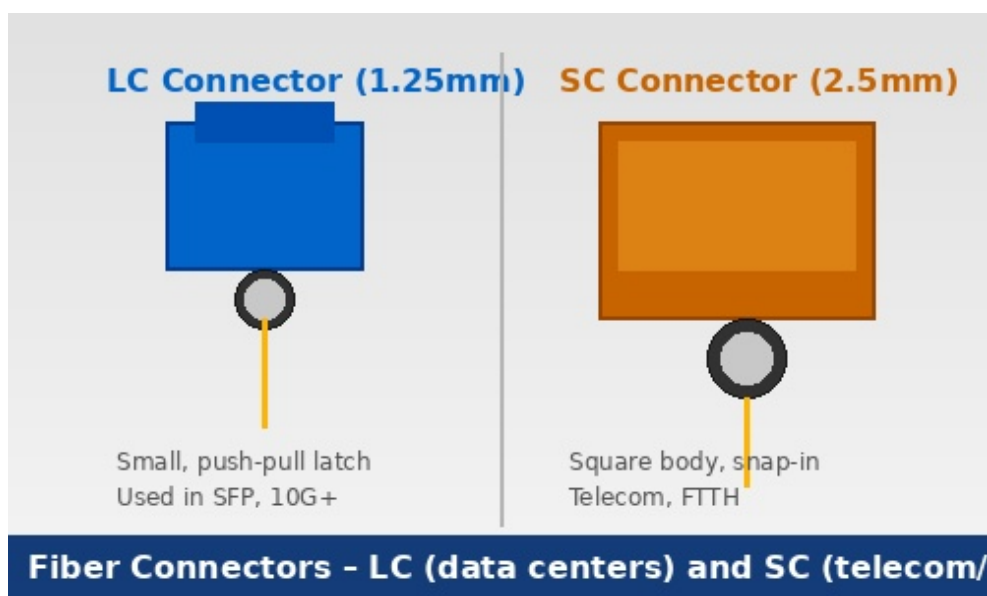


Fig 5: LC Connector (left, small push-pull) and SC Connector (right, square snap-in)

Connector	Ferrule Size	Latch Type	Where Used	Memory Tip
LC	1.25 mm	Push-pull	SFP modules, data centres, 10G+	LC = Little Connector
SC	2.5 mm	Snap in/out	FTTH, telecom, older 1G gear	SC = Square Connector

ST	2.5 mm	Bayonet twist	Legacy campus networks, CCTV	ST = Stab & Twist
FC	2.5 mm	Screw-on	Test equipment, telecom	FC = Fixed Connector
MPO/MTP	12 or 24 fibre	Push-pull	40G/100G/400G parallel links	MPO = Many fibres at once

■ **Tip:** LC connectors are the most common today — almost all SFP modules use LC connectors!

Chapter 7: SFP Transceiver Modules

An **SFP (Small Form-factor Pluggable)** module is a small hot-swappable device that you plug into a network switch or router to add a port. It converts electrical signals from the switch into optical signals (light) for the fiber cable — or copper signals for RJ-45.

Think of it like an **adaptor** — the same switch port can support fiber or copper, short or long distance, just by changing the SFP module!

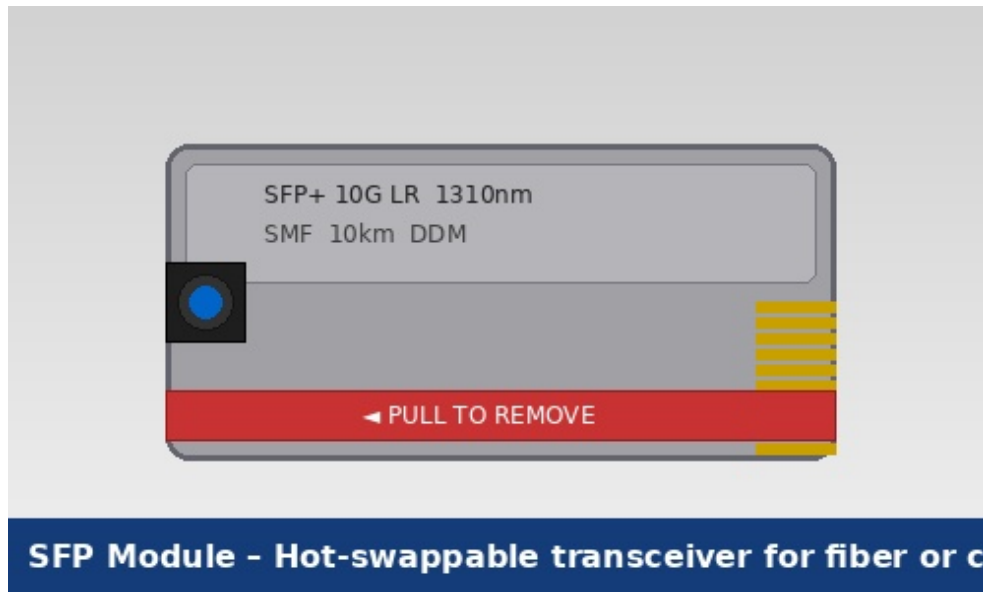


Fig 6: SFP Module – plugs into switch, LC fiber port on the left, gold contacts on the right

7.1 SFP Module Types — From Slow to Very Fast

Module	Speed	Connector	Max Distance	Use Case
SFP	1 Gbps	LC or RJ45	Up to 80 km	1G links, access layer
SFP+	10 Gbps	LC	Up to 80 km	Standard 10G data centre
SFP28	25 Gbps	LC	Up to 10 km	Server uplinks, 25G
QSFP+	40 Gbps	MPO-12	Up to 10 km	40G aggregation links
QSFP28	100 Gbps	MPO-12	Up to 40 km	100G core/data centre
QSFP-DD	400 Gbps	MPO-16	Up to 2 km	400G ultra high speed

7.2 Common SFP Sub-Types by Distance

- **SX (Short Range):** Uses 850 nm light. Works with MMF (OM3/OM4). Range: 300–550 m
- **LX / LR (Long Range):** Uses 1310 nm laser. Works with SMF. Range: 10 km
- **ER (Extended Range):** Uses 1550 nm laser. Works with SMF. Range: 40 km
- **ZX (Ultra Long):** Uses 1550 nm. SMF only. Range: 80 km
- **-T (Copper):** Uses RJ-45 instead of fiber. Range: 100 m. Good for short links

- **BiDi (Bidirectional):** Both directions on ONE fiber using two wavelengths

■ **Tip:** Always check the SFP's wavelength matches your fiber type! SR = multimode; LR/ER/ZX = single-mode.

Chapter 8: Quick Compatibility Reference

8.1 SFP Module vs Cable Type – At a Glance

SFP Module	Speed	Copper Cat5e+	OM3/OM4 Fibre	Single-Mode
SFP-SX	1G	No	Yes (550 m)	No
SFP-LX	1G	No	No	Yes (10 km)
SFP-T (Copper)	1G	Yes (100 m)	No	No
SFP+-SR	10G	No	Yes (300 m)	No
SFP+-LR	10G	No	No	Yes (10 km)
SFP28-SR	25G	No	Yes (100 m)	No
QSFP28-SR4	100G	No	Yes (100 m)	No
QSFP28-LR4	100G	No	No	Yes (10 km)

8.2 Choosing the Right Cable — Decision Guide

My Situation	Use This
Home WiFi router to PC, up to 1G	Cat 5e, straight-through, RJ-45
Office upgrade, want 10G future-proof	Cat 6a, 10G switch, RJ-45
Factory floor with lots of electrical noise	Cat 6a STP or Cat 7, shielded RJ-45
Server rack inside data centre (< 30 m)	Cat 8 or DAC copper + SFP28/QSFP28
Between floors in same building (< 300 m)	OM3/OM4 fibre + LC + SFP+-SR
Between buildings on campus (< 10 km)	Single-mode + LC + SFP+-LR
City-to-city or long WAN link (up to 80 km)	Single-mode + LC + SFP-ZX or EDFA
100G data centre core link	OM4 MPO-12 + QSFP28-SR4

■ **Tip:** When in doubt: Cat 6a for copper inside buildings, OM4 multimode for short fibre runs, single-mode SMF for anything over 500 metres!

Summary — Quick Cheat Sheet

Topic	Key Facts to Remember
UTP Cable	4 twisted pairs, PVC jacket, RJ-45 plug, max 100 m
STP Cable	Like UTP but with metal shield — use in high-interference areas
Coaxial Cable	Old technology — used for cable TV/CCTV, NOT modern Ethernet
Cat 5e	1 Gbps, 100 MHz, 100 m — home/office standard
Cat 6a	10 Gbps, 500 MHz, 100 m — recommended new install
Cat 8	40 Gbps, 2000 MHz, 30 m — server rack use only
Single-Mode Fiber	Yellow jacket, 9µm core, laser, up to 80 km, expensive
Multi-Mode Fiber	Orange/Aqua jacket, 50µm core, LED, up to 400 m, cheaper
LC Connector	1.25 mm ferrule, push-pull — standard for SFP/10G+
SC Connector	2.5 mm ferrule, square snap — telecom/FTTH
SFP (1G)	1 Gbps module, LC or RJ-45, works with SMF or MMF
SFP+ (10G)	10 Gbps, LC connector, SR=MMF, LR=SMF
QSFP28 (100G)	100 Gbps, MPO-12, SR4=MMF, LR4=SMF
T568B	Most common wiring standard for straight-through cables

*This guide was prepared by **Rushwik** as part of the Network Instructor training series. All diagrams are for educational purposes.*